Complete Model Progression: From Character Recognition to Paragraph OCR

1. MLP + EMNIST (Baseline)

Task: Single character recognition
Data: EMNIST - 28×28 grayscale characters (814K samples, 62 classes)
Architecture: 3 fully-connected layers (784 → 1024 → 128 → 83)
Parameters: 945,739
Output: Single class prediction per image

Strengths Fast, simple, interpretable

Weaknesses No spatial structure, high param count, can't do sequences

Use: Baseline comparison

2. CNN + EMNIST (Spatial Structure)

Task: Single character recognition with spatial awareness
Data: Same as MLP (28×28 EMNIST characters)
Architecture: 2 ConvBlocks + MaxPool + FC layers
Parameters: 1,654,035
Output: Single class prediction with learned convolution filters

Improvement over MLP Preserves 2D locality, learns hierarchical features, better accuracy

Remaining issue Still single class per image—can't do sequences!

3. LineCNNSimple + EMNIST Lines (Simple Sequences)

Task: Text line recognition (perfectly separated characters)
Data: EMNIST Lines - variable width lines of concatenated 28×28 chars
Architecture: Sliding window CNN, reuse same CNN for each window
Parameters: 1,654,035 (same CNN, applied multiple times)
Output: Sequence of character classes $(B, 83, S)$

Improvement over CNN No fixed size—can handle variable lengths, outputs sequences

Limitation Assumes clean character separation, each window independent (no context)

4. LineCNN + IAM Lines (Hierarchical Features + Context)

Task: Real handwritten line recognition (cursive, overlapping)
Data: IAM Lines - 56×1536 real handwritten text lines (9,862 lines)
Architecture: 9-layer CNN with 3 stride-2 downsampling stages + implicit sliding
Parameters: 1,199,140
Output: Sequence of character logits $(B, 83, S_x)$ where $S_x \approx 48$

Improvement over LineCNNSimple Hierarchical multiscale learning, implicit context, handles real handwriting

Issue No language model—each character predicted independently, can't correct errors

5. LineCNNTransformer + IAM Lines (Add Language Model)

Task: Line recognition with language understanding
Data: Same IAM Lines
Architecture: LineCNN encoder (1.2M params) + Transformer decoder (4 layers, 2.1M params)
Parameters: 3,340,919
Output: Autoregressive token sequence; teacher forcing in training

Improvement over LineCNN Can use context from previous tokens, corrects errors, teacher forcing during training

Tradeoff Autoregressive inference is SLOW (×50 slower than LineCNN)

6. ResnetTransformer + IAM Paragraphs (Full Documents)

Task: Full paragraph OCR with multi-line text
Data: IAM Paragraphs - 576×640 full paragraph images (multiple lines)
Architecture: ResNet-18 encoder (11M params, pre-trained) + Transformer decoder (4.3M params)
Parameters: 15,679,796 (or 4.5M trainable if ResNet frozen)
Output: Autoregressive sequence with newline tokens (up to 682 chars)

Improvement over LineCNNTransformer

- Handles 2D layout naturally (paragraphs, not just lines)
- Pre-trained ResNet provides strong features
- Deeper architecture (18 layers) learns richer representations
- Skip connections enable optimization in very deep nets
- Explicit newline token models document structure

Cost 15-30MB model, requires GPU, slow inference (1-5s per page)

Parameter Comparison

Model	MLP	CNN	LineCNNSimple	LineCNN	LineCNNTf	ResNetTf
Parameters (M)	0.95	1.65	1.65	1.20	3.34	15.68
Model Size (MB)	3.6	6.3	6.3	4.6	12.8	60
Inference speed (Hz)	1000+	500+	400+	300+	5-10	1-2
Max input size	28×28	28×28	28×Var	56×1536	56×1536	576×640
Dataset	EMNIST	EMNIST	EMNIST-L	IAM-L	IAM-L	IAM-P
Typical Accuracy	82%	94%	89%	93%	96%	92%

Key Insights

Architecture Evolution

- MLP → CNN: Add spatial structure (2D convolution)
- CNN → LineCNNSimple: Make sequence output (sliding window)
- LineCNNSimple → LineCNN: Hierarchical multiscale learning
- LineCNN → Transformer: Add language model (context)
- Line → Paragraph: Switch to 2D layout (ResNet + Transformer)

Dataset Progression

- EMNIST: Small (28×28), perfectly cropped characters
- EMNIST Lines: Medium (28×Var), concatenated characters
- IAM Lines: Large (56×1536), real handwriting, cursive
- IAM Paragraphs: Largest (576×640), full documents with layout

Tradeoff Curve

- As complexity ↑ (params, layers, data), accuracy ↑ but speed ↓
- MLP: 3.6MB, 1000 Hz, 82% acc
- ResNet: 60MB, 1 Hz, 92% acc
- Choose based on accuracy-speed-memory constraints!